

why do I need ***two*** loops.

Doesn't one loop induce a flux through itself?
shouldn't that also lead to an EMF?

Well, yes, and all the same arguments would hold.
(think about what happens if you put loop 1 and 2 on top of
each other)

Thus, I could define the **self-inductance** (or inductance for
short) of a circuit component as

$$\mathcal{E} = -L \frac{dI}{dt}$$

Where $\mathcal{E}$ is the voltage drop across the element and I is the
current flowing through it.



The inductor:

A circuit element where changing the current through it results in a voltage drop that tries to resist that change.

As before, the units of inductance are **Henry[s]**.

Engineering self inductance:

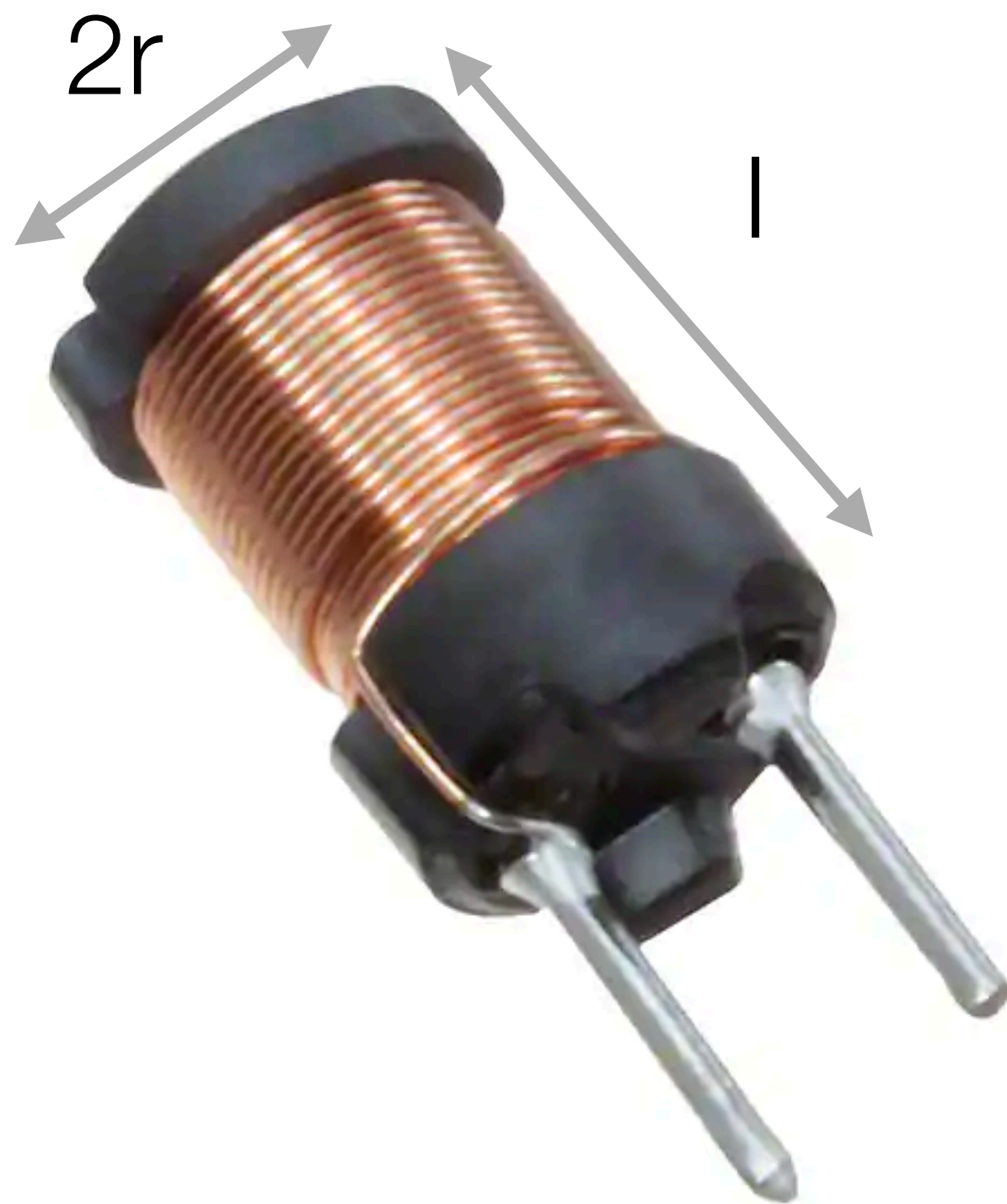
For many geometries, it can be easy to calculate the inductance. This is especially true for geometries with known magnetic fields. If you know the flux per I, then you also know L

$$L = \Phi_M/I$$

If you want to engineer a specific inductance, you just turn this relationship around.



Example: the inductance of a solenoid



pretty much what's in your kit...

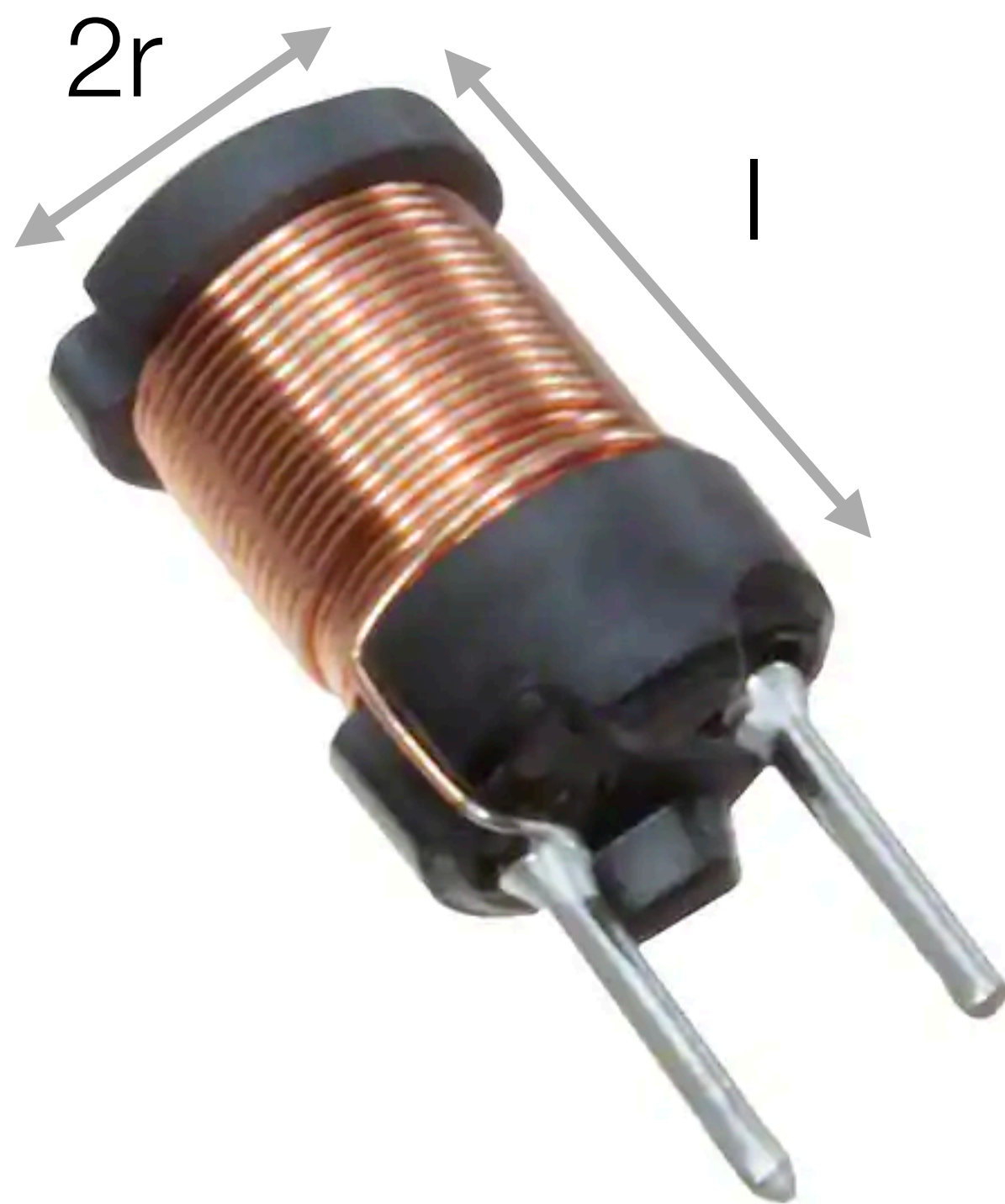
We know the magnitude of the B field generated when a current flows through a solenoid is

$$B = \mu_0 n I$$

Where n is the number of turns per unit length. So the flux through each loop in the solenoid is

$$\Phi_B = \mu_0 n I \pi r^2$$

Example: the inductance of a solenoid



pretty much what's in your kit...

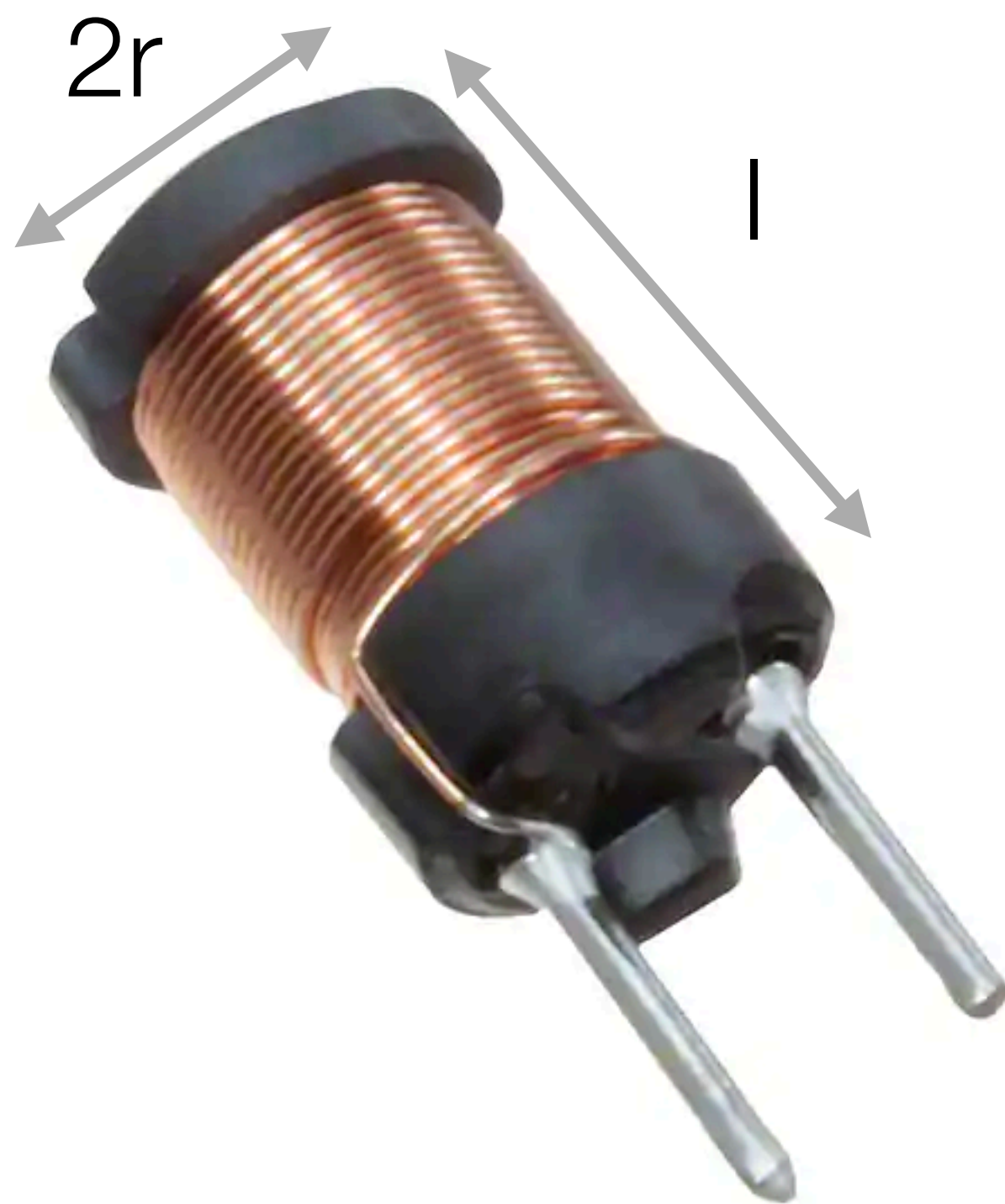
Now if I is made to change, that would induce an EMF around each loop of

$$\mathcal{E}_l = - \frac{d\Phi_B}{dt} = - \mu_0 n \pi r^2 \frac{dI}{dt}$$

And the total drop across all the loops is just the sum of all the individual loop EMFs, or

$$\mathcal{E}_s = - \mu_0 n \pi r^2 (nl) \frac{dI}{dt}$$

Example: the inductance of a solenoid

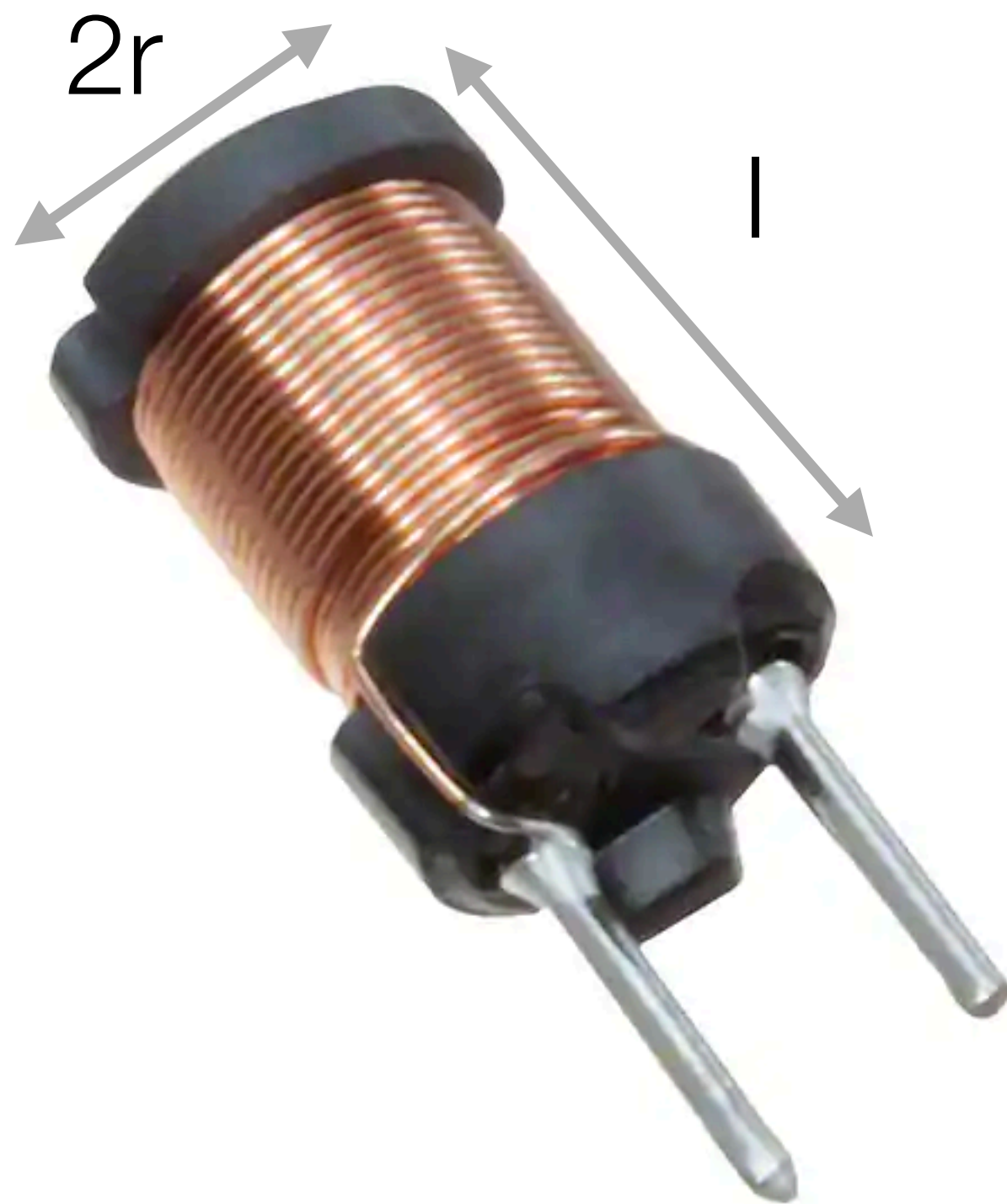


Comparing to the definition of inductance ($\mathcal{E} = -L \frac{dI}{dt}$) we find that if the total voltage drop across the solenoid is $\mathcal{E}_s = -\mu_0 n \pi r^2 (nl) \frac{dI}{dt}$, then the inductance must be

$$L = \mu_0 n^2 \pi r^2 l$$

pretty much what's in your kit...

Example: the inductance of a solenoid



turns density squared

$$L = \mu_0 \overbrace{n^2} \underbrace{\pi r^2 l}_{\text{volume}}$$

Another way of putting things:

$n = l/d$ where d is the diameter of the wire, so

$$L = \mu_0 \pi \frac{r^2}{d^2} l$$

pretty much what's in your kit...

Suppose we fixed the aspect ratio of r/d to some value,

$$L = \mu_0 \pi \frac{r^2}{d^2} l$$

Then we get a **proportionality** between the **inductance** and the **length** of the solenoid.

So how big do you have to make your solenoid to get an inductance of.....



The inductor as a circuit element.

As usual, let's see what happens when you put inductors in series or in parallel?



Say I take two Inductors and put them in series

Any current through the first one must equal the current through the second ($I_1 = I_2 = I$)

Likewise the total voltage drop across is given by the sum

$$\mathcal{E}_T = \mathcal{E}_1 + \mathcal{E}_2$$

So, using the inductance equation

$$\mathcal{E}_T = -L_1 \frac{dI_1}{dt} + -L_2 \frac{dI_2}{dt} = -(L_1 + L_2) \frac{dI}{dt}$$



Say I take two Inductors and put them in series

So the two inductors in series could be replaced with a single inductor whose inductance equals their sum.

$$\mathcal{E}_T = = - (L_1 + L_2) \frac{dI}{dt} = - L_{eff} \frac{dI}{dt}$$

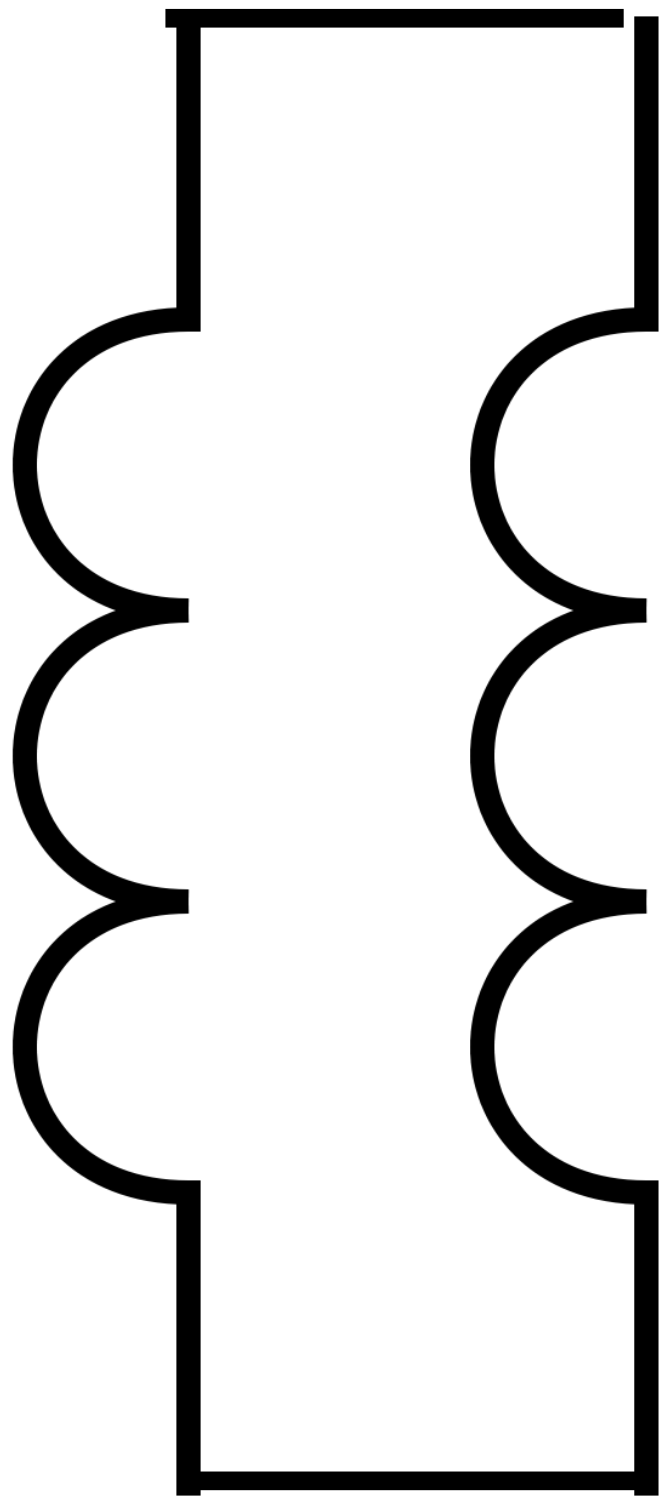
What about parallel?

Here the voltage drop across both must be the same
(they're connected by a wire)

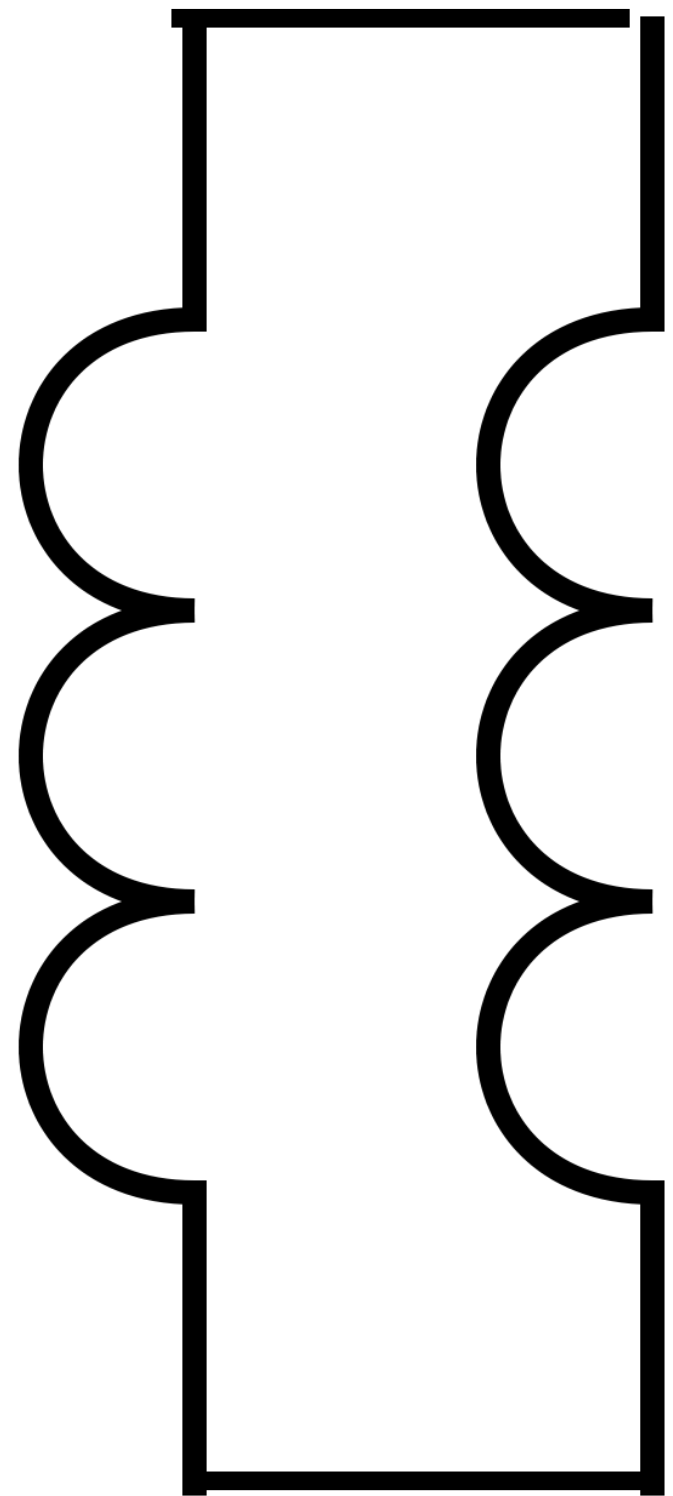
$$\mathcal{E} = -L_1 \frac{dI_1}{dt} = -L_2 \frac{dI_2}{dt}$$

but the current flowing into the circuit has two different
contributions $I = I_1 + I_2$, giving

$$\frac{dI}{dt} = \frac{dI_1}{dt} + \frac{dI_2}{dt} = -\mathcal{E} \left(\frac{1}{L_1} + \frac{1}{L_2} \right)$$



What about parallel?



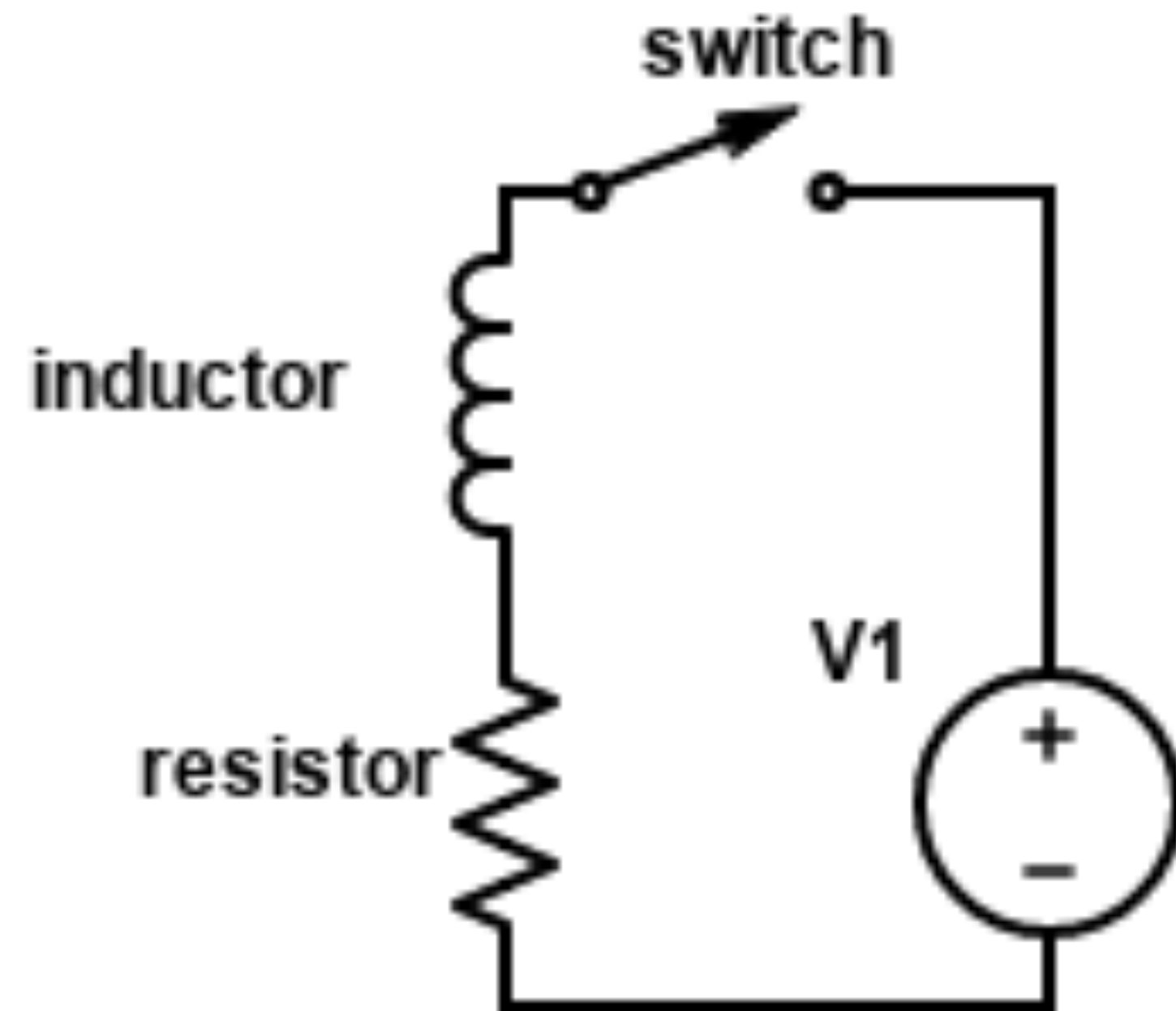
So we find now that the circuit behaves like an effective inductor of inductance

$$\frac{dI}{dt} = \frac{dI_1}{dt} + \frac{dI_2}{dt} = -\mathcal{E}\left(\frac{1}{L_1} + \frac{1}{L_2}\right)$$

or solving for the effective inductance

$$L_{eff} = 1/(1/L_1 + 1/L_2)$$

Ok that's fine, but what does an inductor **do** when you put it in a circuit?



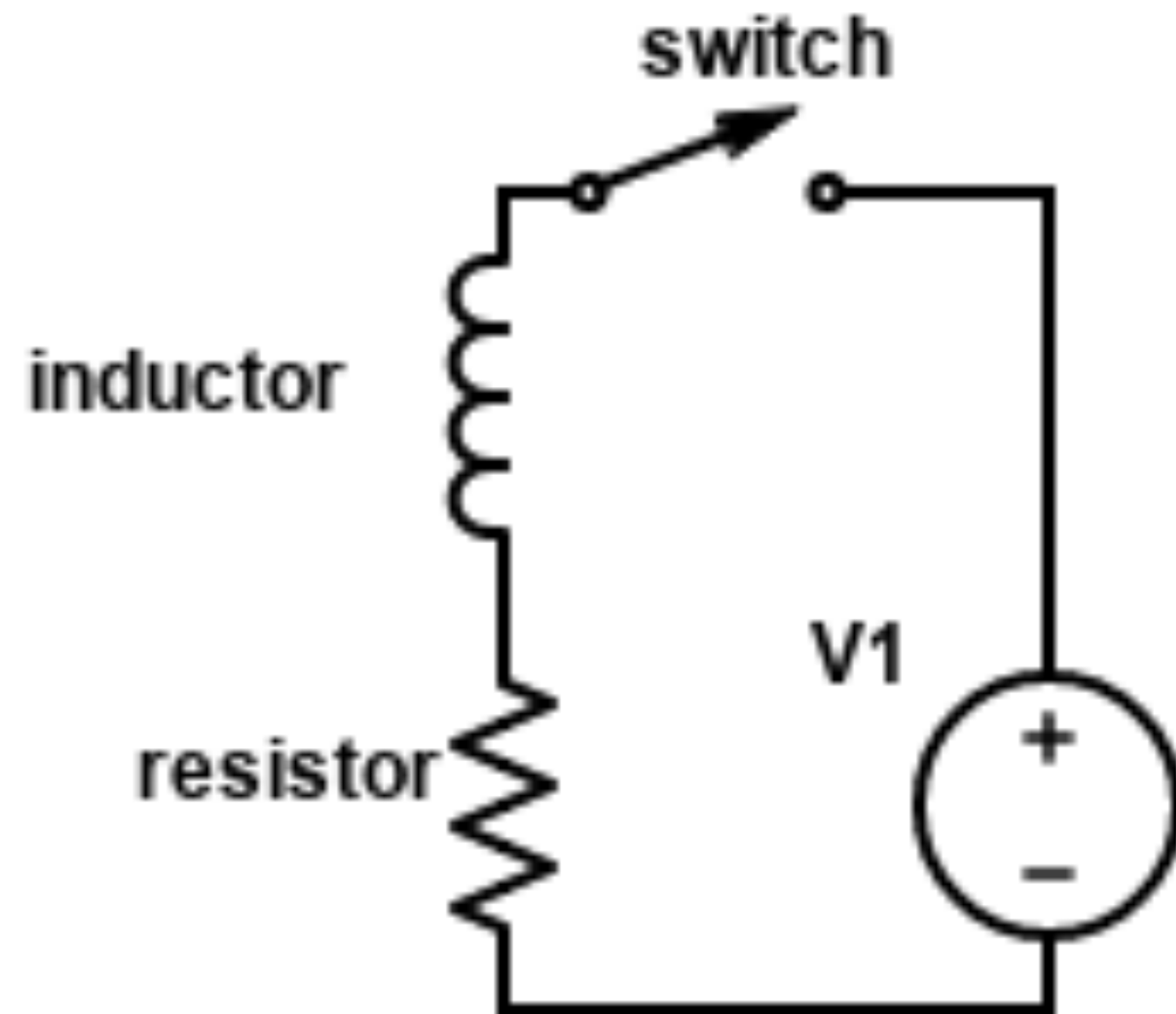
Wire a resistor and inductor in series.
now hook them up to a battery through a switch.
When the switch gets closed, what happens?

Inductors fight changes in current.

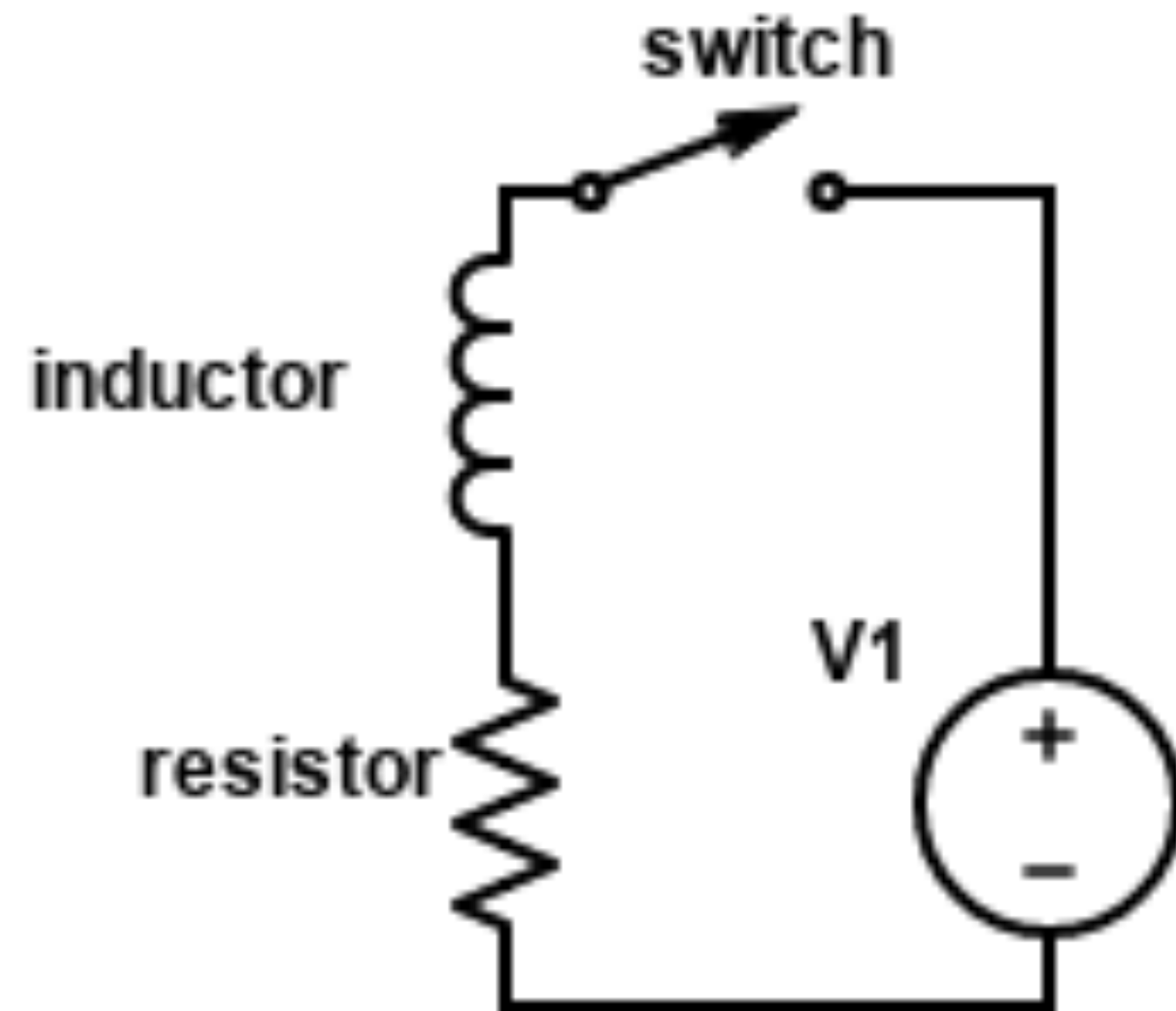
When the switch is open, there's no current.
When it closes, the current wants to go up from zero, but the inductor applies a back EMF to resist this change.

Indeed early on, the resistor plays little role since initially no current is flowing.

It's only at long times, when the system is settling into an equilibrium that I can get a major effect from the resistance.



If we measured the voltage drop across the resistor, that still has to equal the combined drop across the source and inductor.



The drop along the resistor is still $V_r = IR$

While the total drop across the battery and inductor is $V_b + \mathcal{E}_L = V_b - L \frac{dI}{dt}$

So altogether, we get

$$IR = V_b - L \frac{dI}{dt}$$

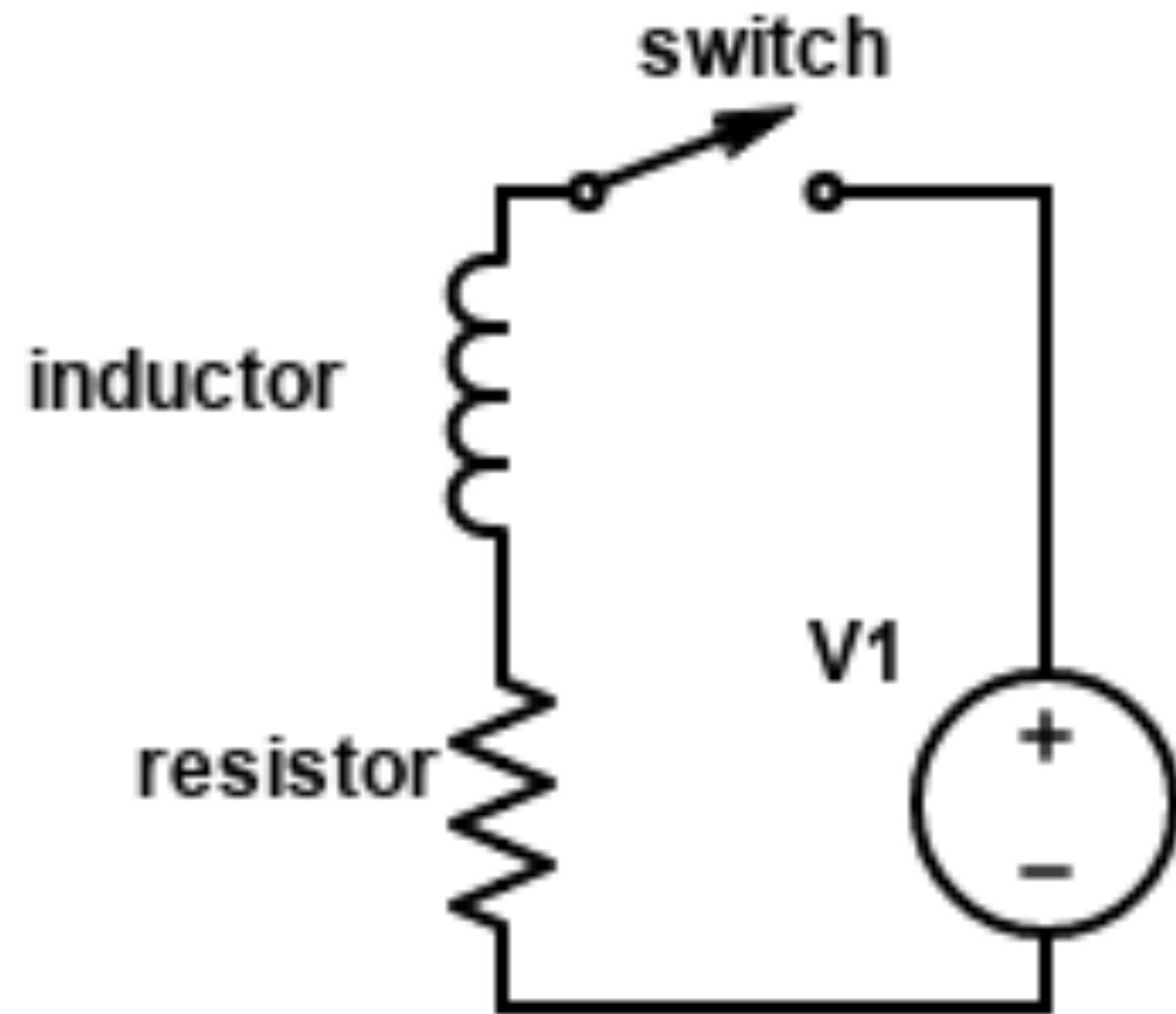
Now if we look at

$$IR = V_b - L \frac{dI}{dt}$$

It says two things.

Early on $I \sim 0$, so $V_b \approx L \frac{dI}{dt}$ and the inductor rules

At long times $\frac{dI}{dt} \approx 0$, so we get $V_b \approx IR$ which
is basically just the resistor.



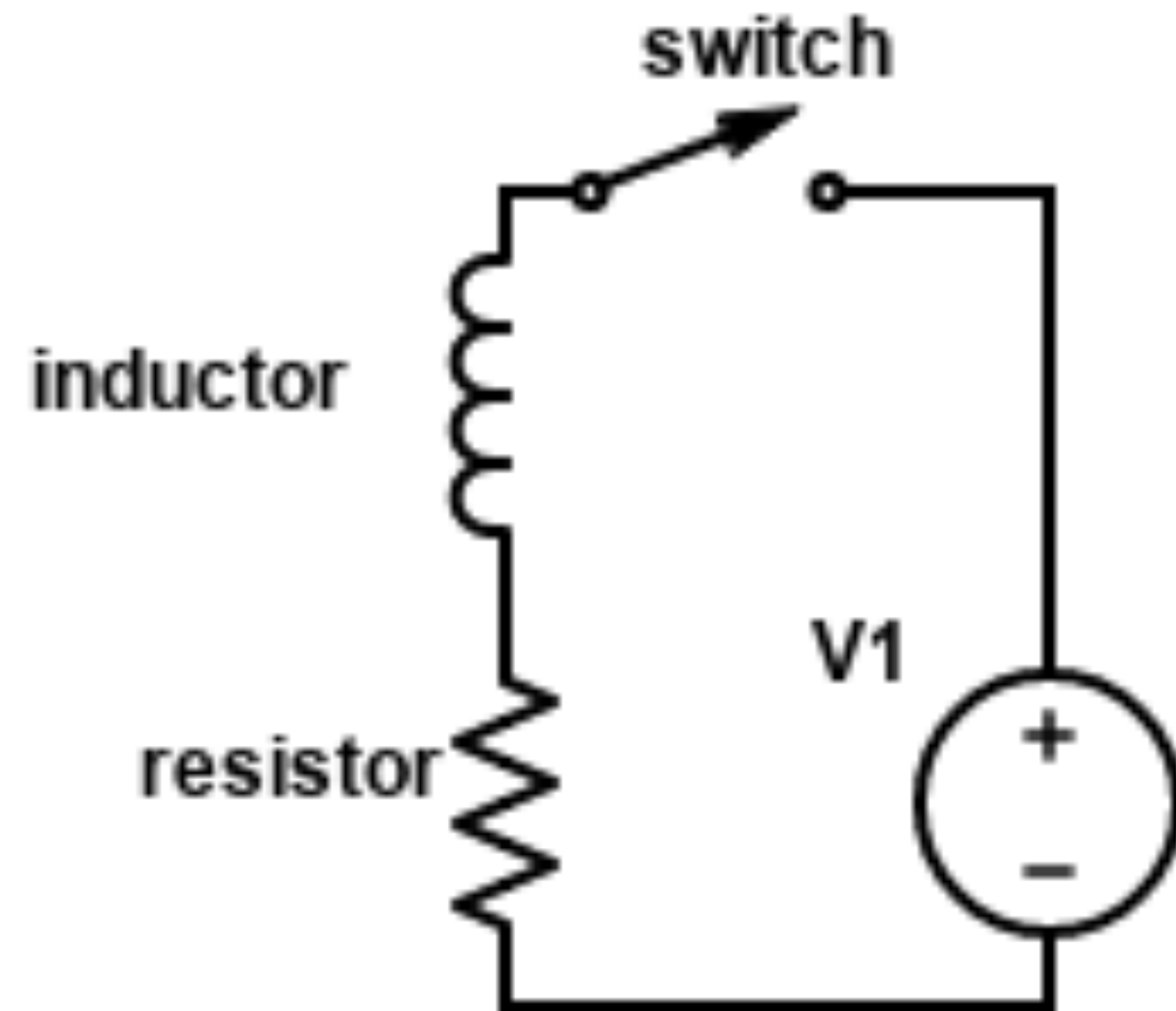
What happens in between?

$$IR = V_b - L \frac{dI}{dt}$$

As with the RC circuit, we have the transient behavior connecting the two is described by a function that's proportional to its derivative plus a constant. So let's make a guess for the solution of

$$I(t) = \frac{V_b}{R} (1 - \exp[-t/\tau])$$

Would asymptote to the resistor regime at long times and goes to zero as t goes to zero



Plugging the trial solution of

$$I(t) = \frac{V_b}{R}(1 - \exp[-t/\tau])$$

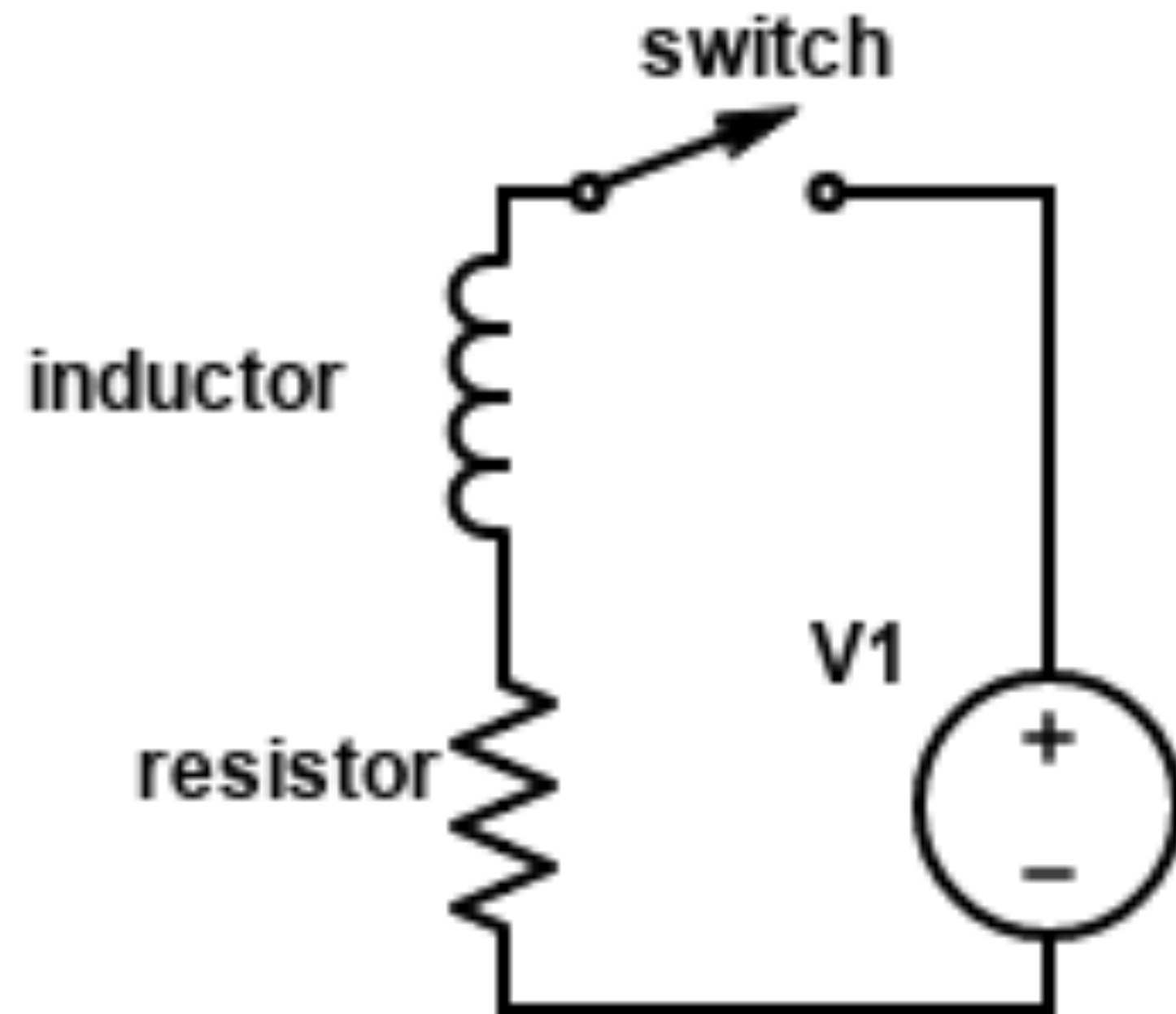
Into the equation for the circuit dynamics,

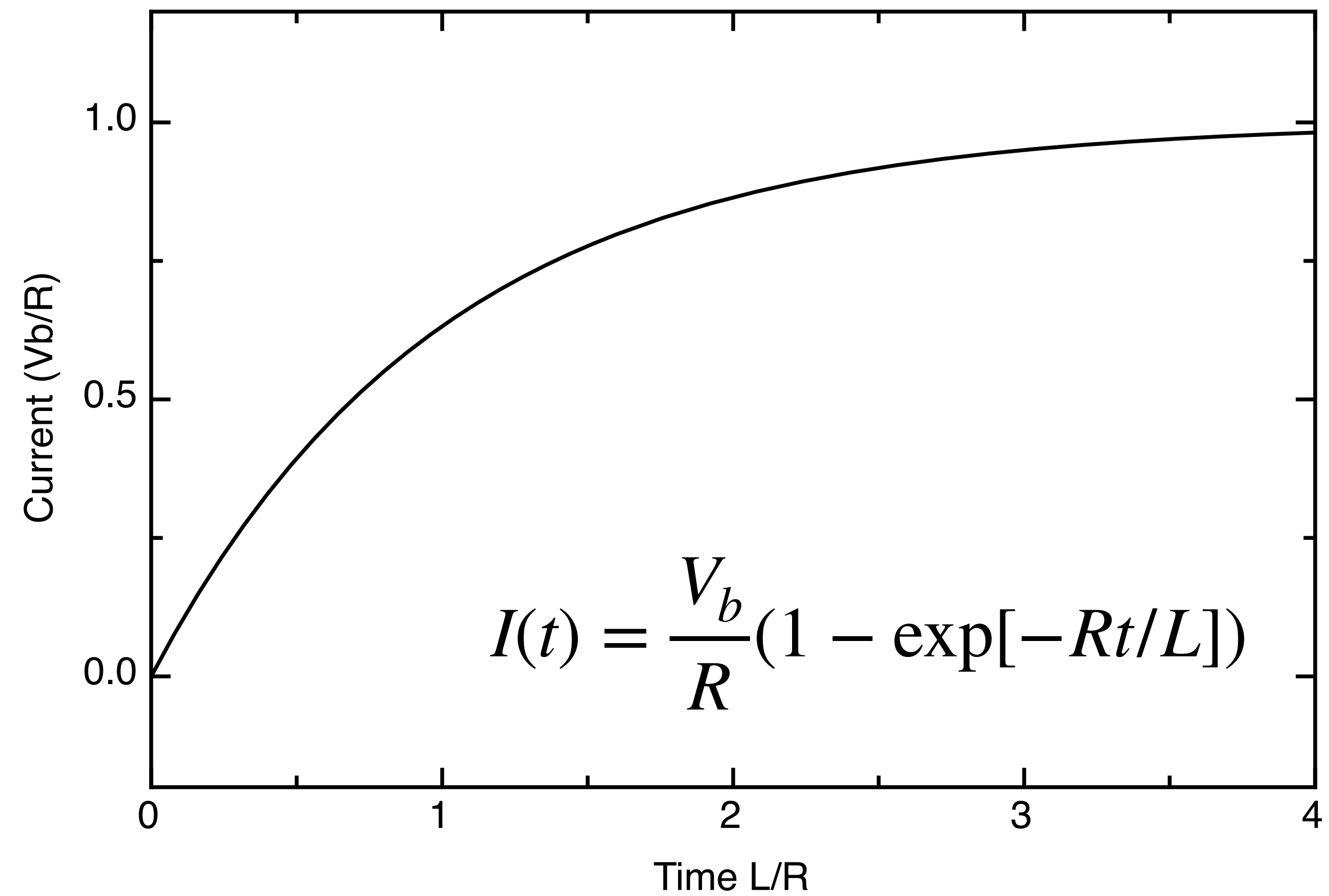
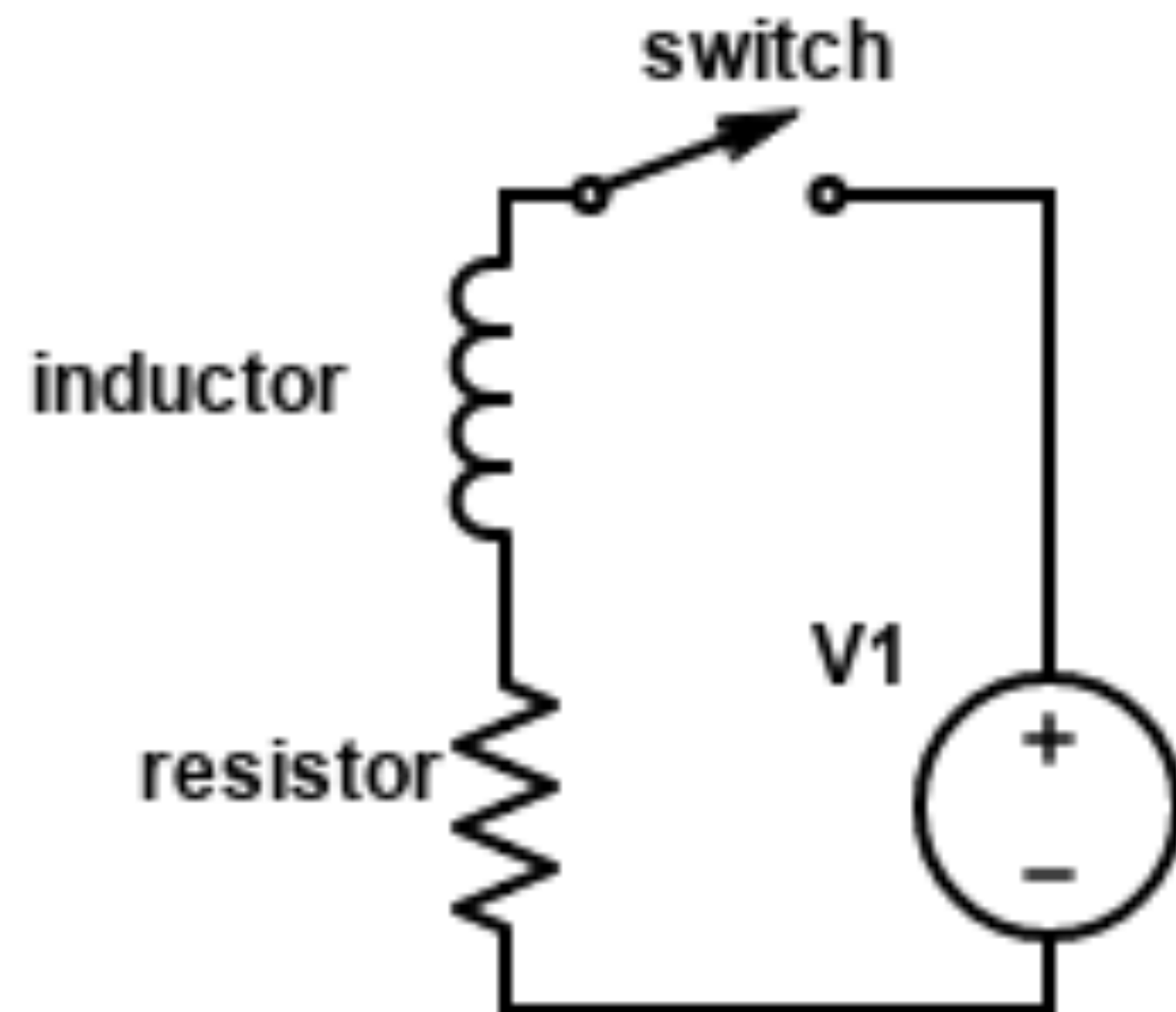
$$IR = V_b - L \frac{dI}{dt}$$

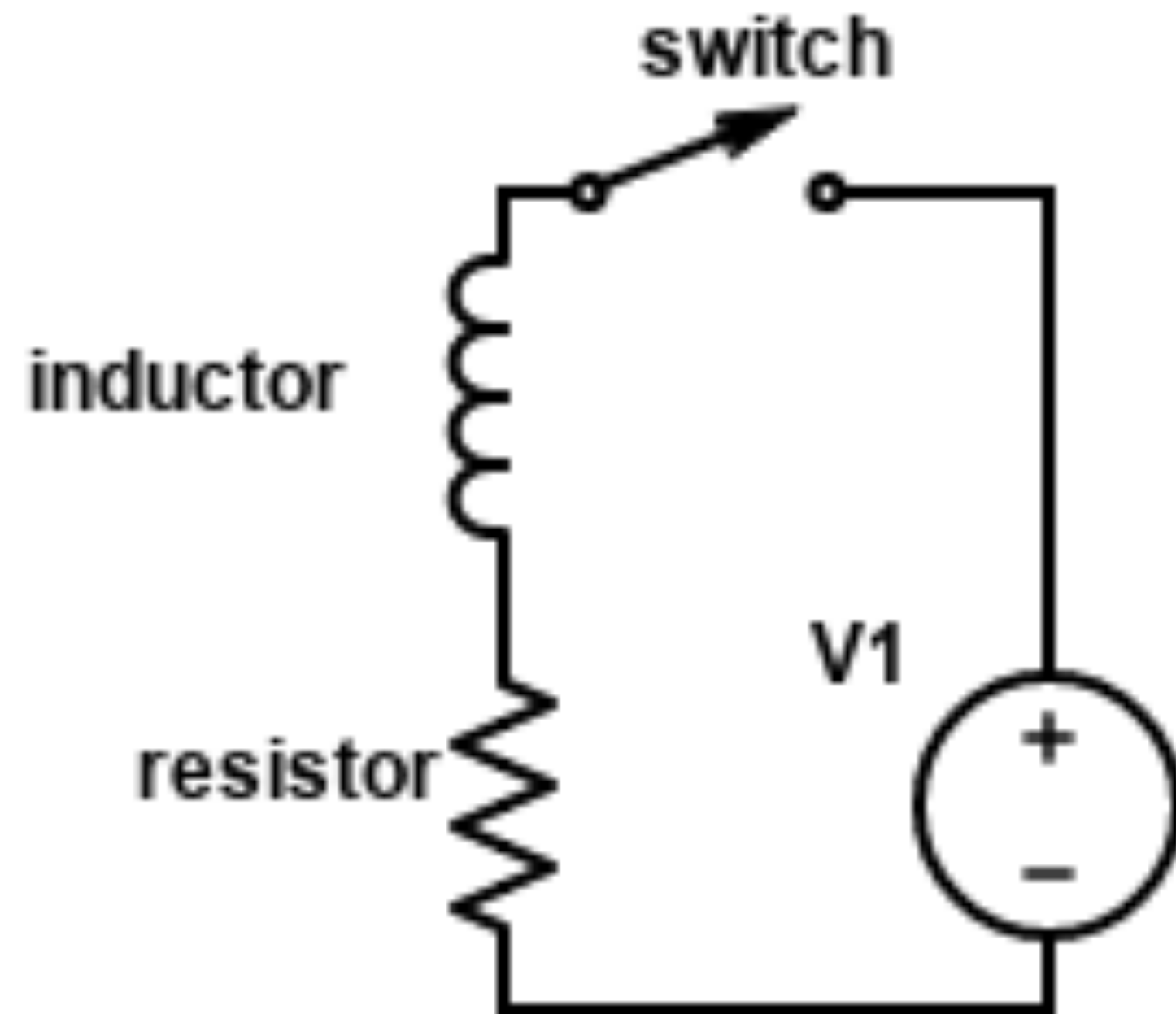
we get that we have a solution if

$$\tau = L/R.$$

Same kind of exponential relaxation we had in RC,
but this time with current instead of *voltage*!

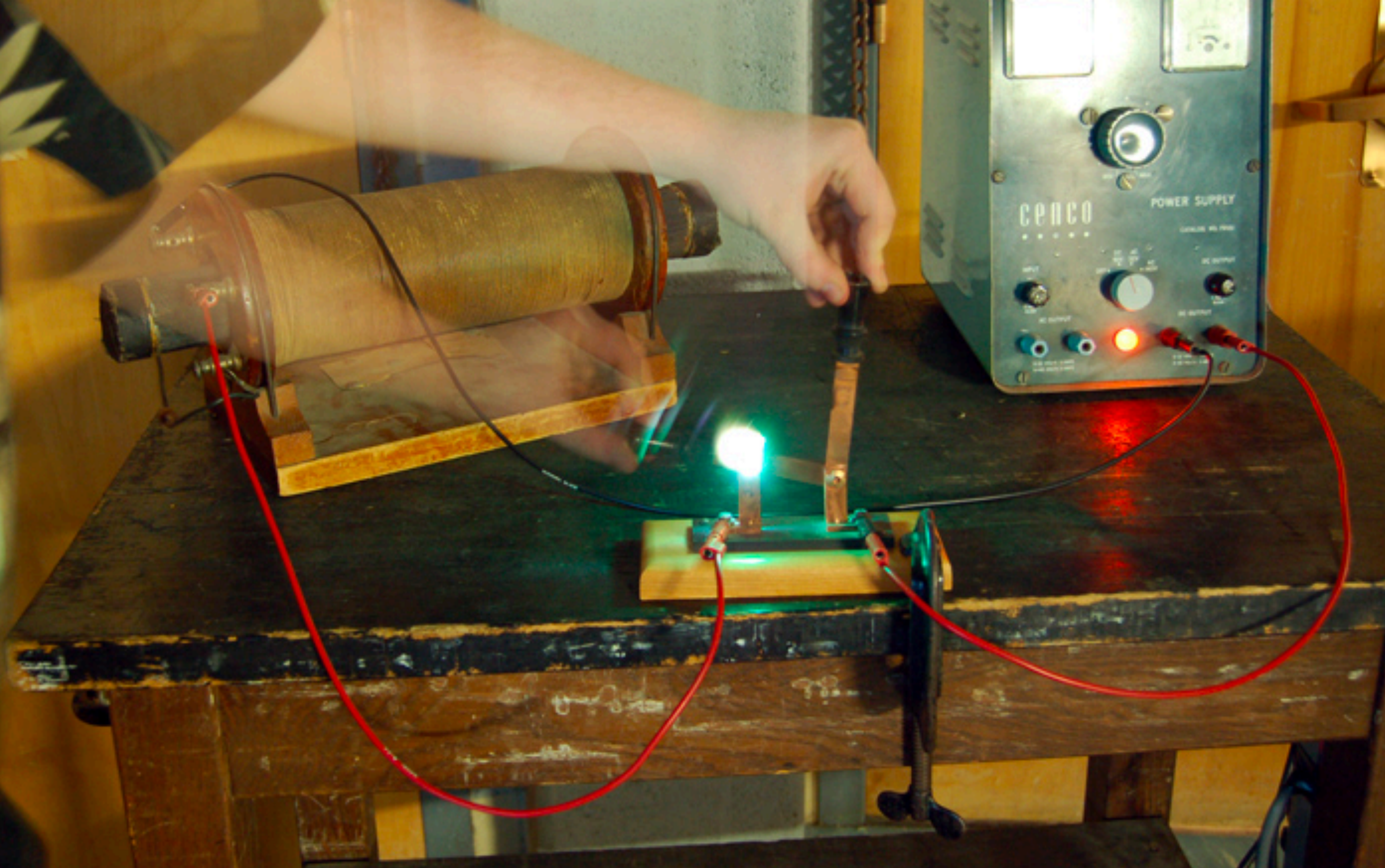


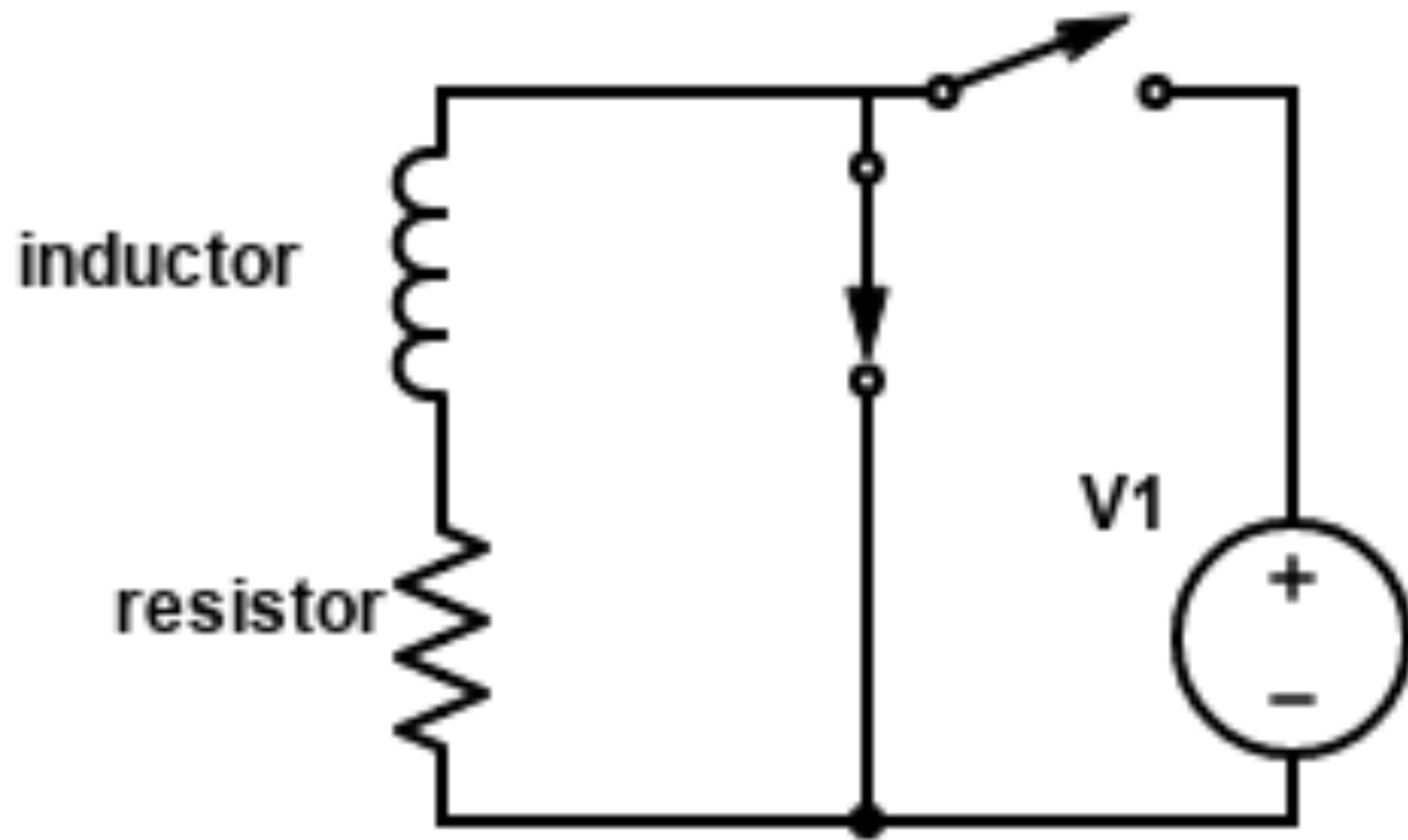




Ok so now that the current I is happily flowing, what **would happen if you opened the switch again?**

What would the voltage drop across the inductor and resistor be?





Lets do something safer:

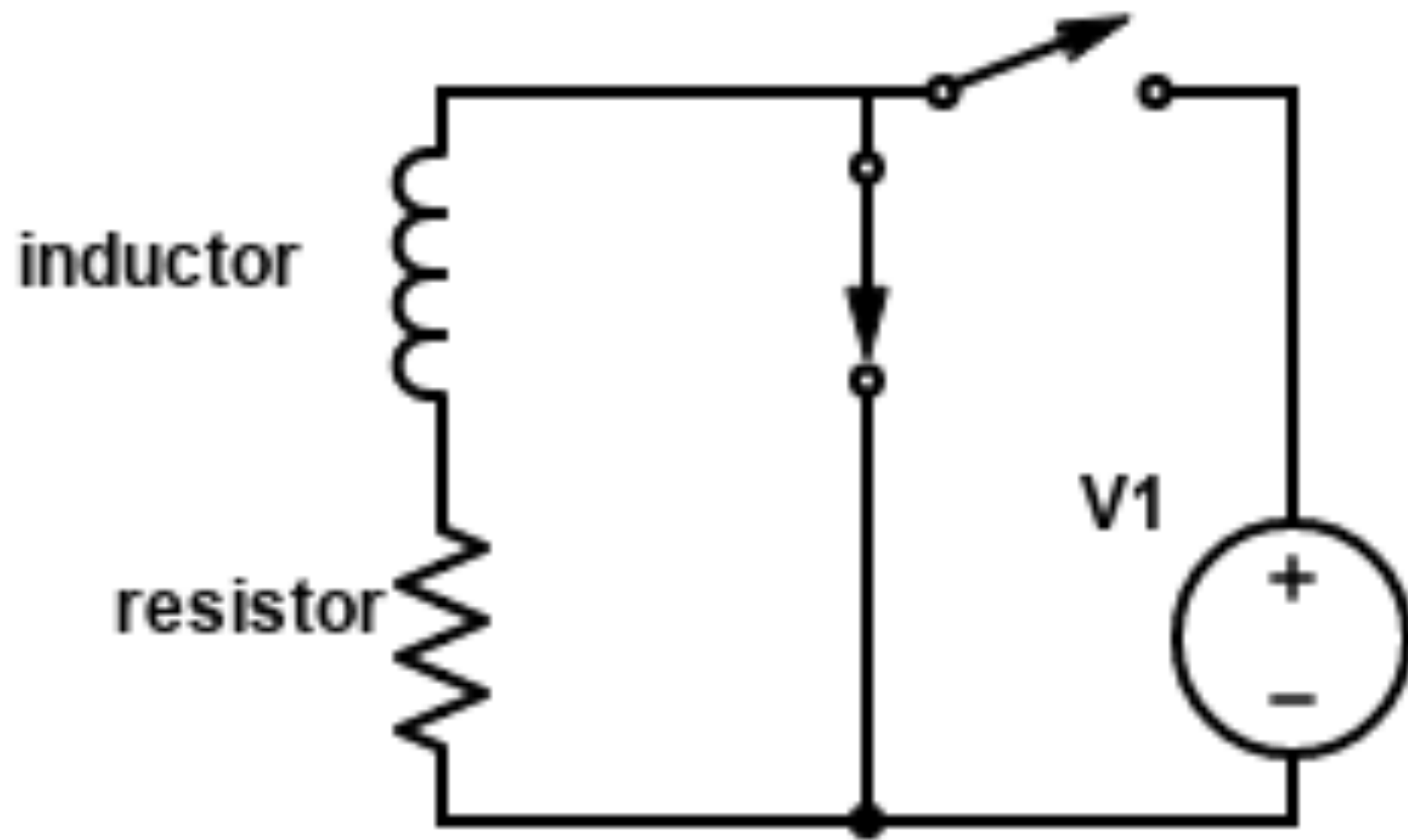
say we short the RL combination and disconnect the battery at the same time

Then we get the RL sub circuit satisfies

$$L \frac{dI}{dt} + RI = 0$$

Which again is solved by

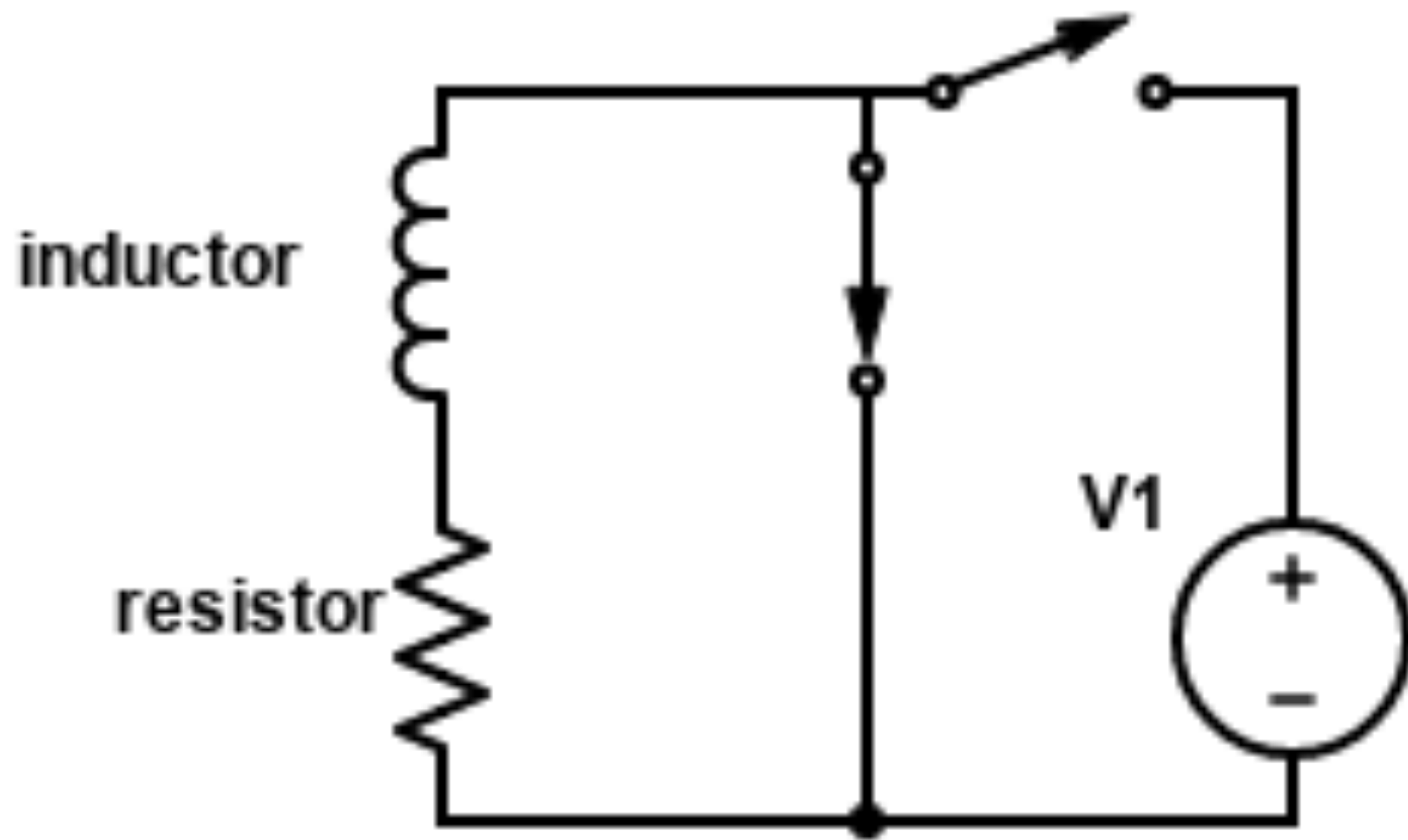
$I = I_0 \exp[-Rt/L]$ where I_0 is the initial current.



Energy from the magnetic field:

Note that current is still flowing through the resistor. That current is being driven by the EMF across the inductor. In effect, **there was energy stored up in the inductor by the initially moving current that's now being extracted.**

Lets work out how much.



We said the current in the circuit was

$$I = I_0 \exp[-Rt/L]$$

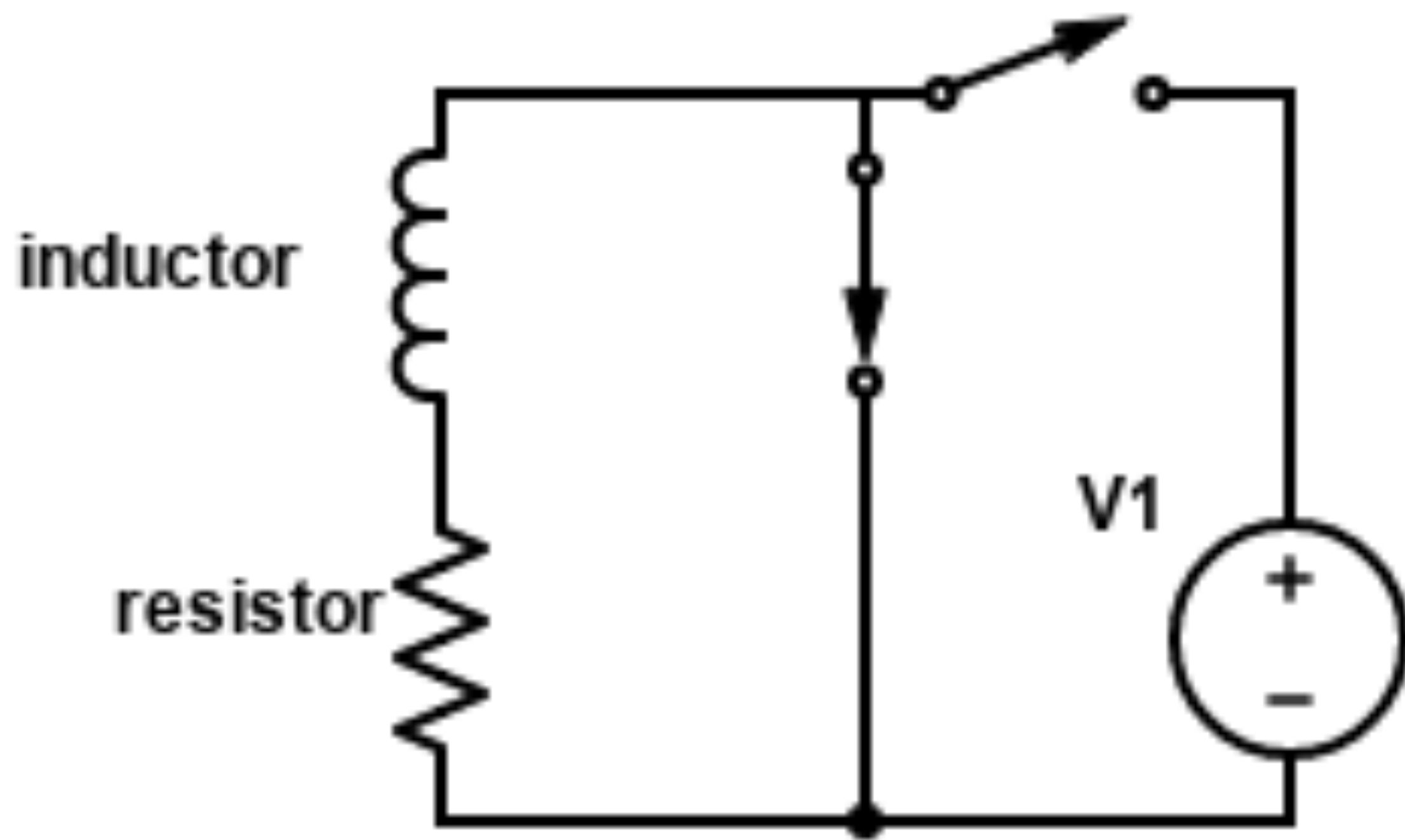
and the power burned moving current through a resistor is

$$P = RI^2$$

so the total energy moved from the field to the flowing charge in the resistor is

$$U = \int_0^\infty dt RI^2 = \int_0^\infty dt RI_0^2 \exp[-2Rt/L] = -RI_0^2(L/2R) \exp[-2Rt/L] \Big|_0^\infty$$

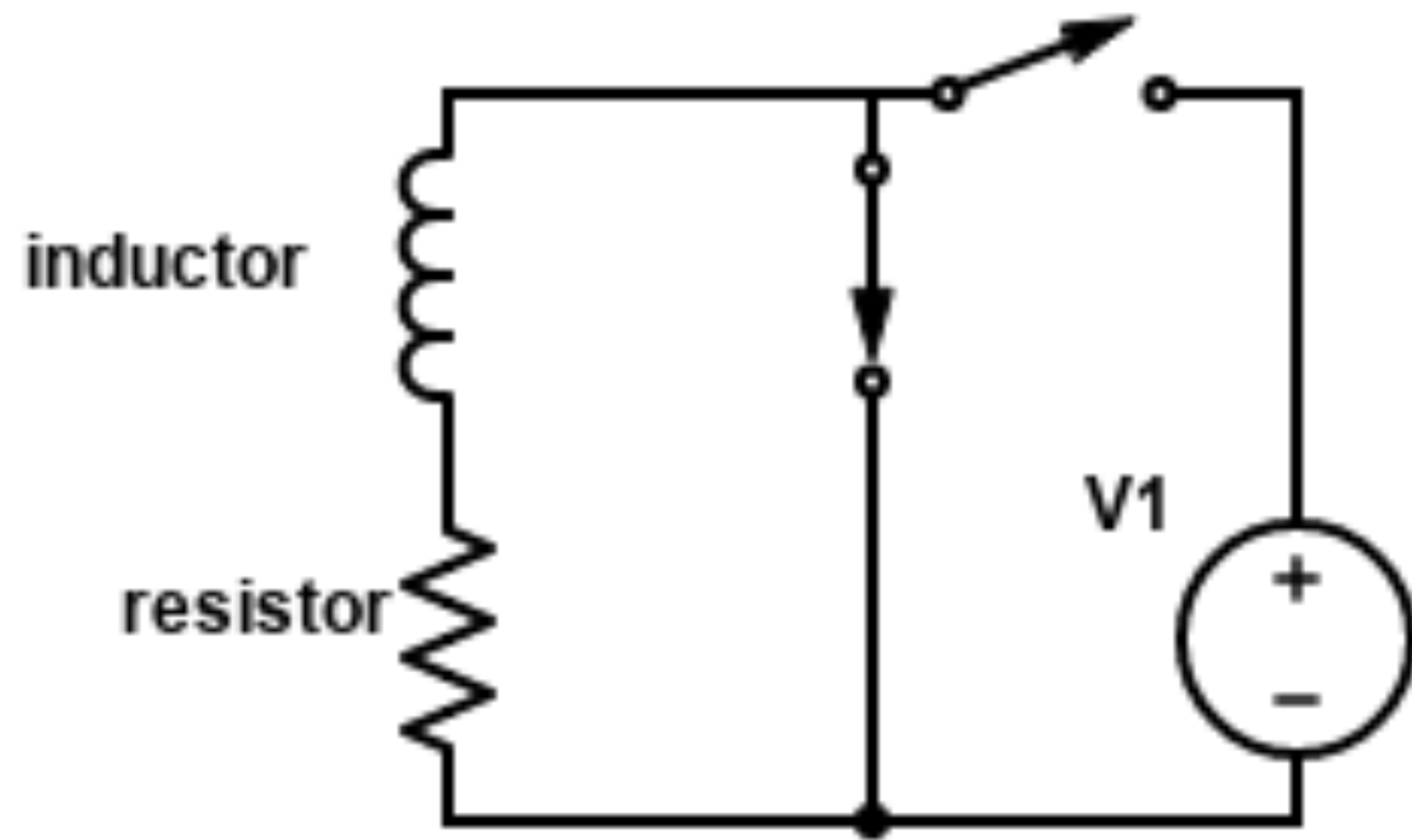
$$U = \frac{1}{2} LI_0^2$$



Thus, we find that

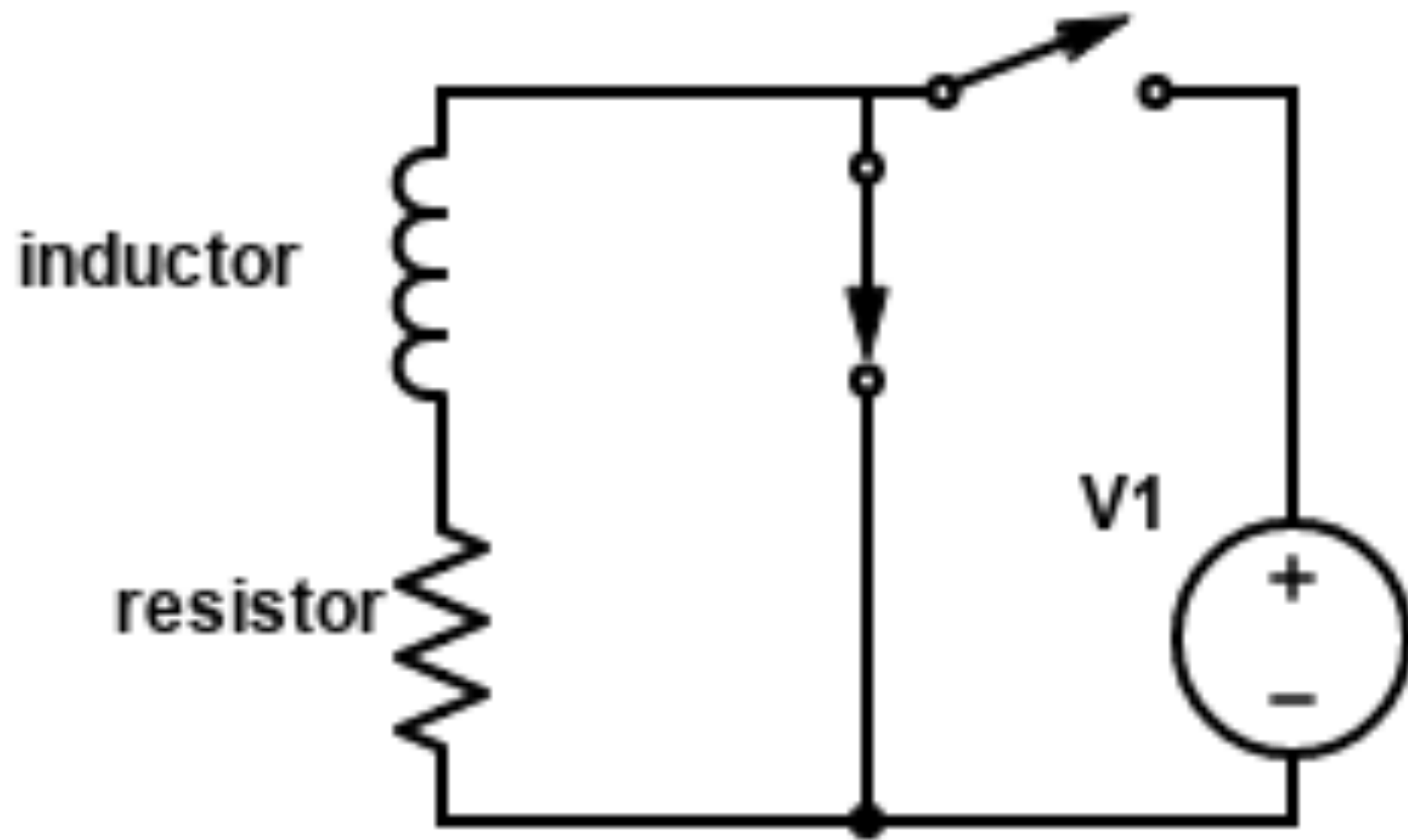
$$U = \frac{1}{2}LI_0^2$$

Note: the source of this energy is essentially the decaying magnetic field. That decaying B is being converted to an electrical field via induction, and driving the charges around the loop.



How did that energy get in there in the first place?

The battery! That's what set the field up in the first place



Consistency check:

the work done by the battery better match the energy stored in the field.

Recall the EMF is the work per unit charge. So to push a charge $I dt$ through an inductor, a battery needs to do a work of $dW = - \mathcal{E} I dt = LI \frac{dI}{dt} dt$

Thus the total energy loaded into the inductor is

$$W = \int dt LI \frac{dI}{dt} = \frac{1}{2} LI_0^2 \text{ which is what the resistor sucked back out.}$$

With induction in the mix,
B can't do work, ***but it can store energy.***

*Creating or destroying a B field requires changing its flux.
This in turn creates an electric field.*

You do work against the induced E to set up a magnetic field.

***You can get that work back from the induced E when you get rid
of a magnetic field.***

***But note, its B that holds onto the energy for you:
theres no E around in the meantime.***